

## EV CHARGING ANALYTICS

### User Guide

**Dashboard:** EV Charging Analytics

**Purpose:** Monitoring Renewable Integration, Energy Demand and Dynamic Pricing Performance

**Data Coverage:** January 2025 – March 2025

---

### 1. Introduction

The **EV Charging Analytics** dashboard is an interactive Power BI analytics application designed to monitor electric vehicle charging activity and operational performance.

The dashboard helps users analyze:

- Charging demand and charging sessions
- Energy consumption
- Charging duration
- Waiting time
- Charging cost
- Vehicle performance
- Charger performance
- Charging station performance
- Location-level demand
- Vehicle-type behavior

The dashboard is designed with interactive navigation, filters, tooltips, drill-through analysis, bookmarks, user information, RLS indicators, and operational notifications.

---

### 2. Opening the Dashboard

#### Step 1 — Sign in

Open the Power BI report shared with you.

Sign in using the Microsoft/organizational account that has been granted access to the report.

If your organization has provided access through your business email, use that account when prompted.

#### Step 2 — Open the Report

After signing in:

1. Open the Power BI workspace or shared report.
2. Select **EV Charging Analytics**.
3. Wait for the dashboard to load.
4. The dashboard opens on the **Executive Overview**.

Access to the report depends on the permissions assigned by the report administrator.

---

### 3. Understanding the Dashboard Layout

The dashboard contains four main analytical sections.

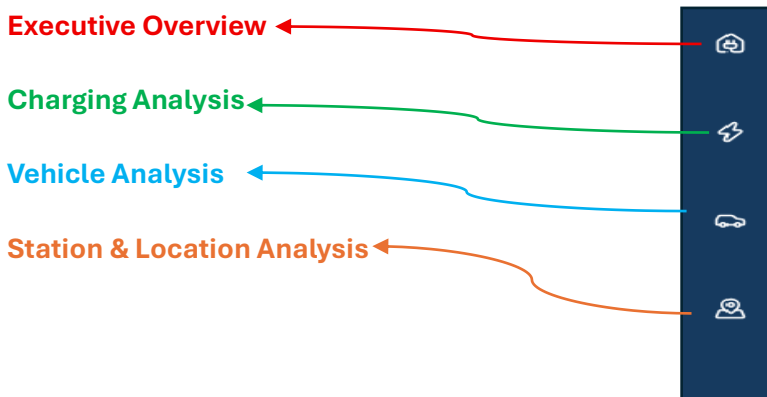
#### Main Sections

1. **Executive Overview**
2. **Charging Analysis**
3. **Vehicle Analysis**
4. **Station & Location Analysis**

These sections can be accessed using the navigation icons on the left side of the dashboard.

The left navigation bar contains icons representing the different analytical views.

---



## 4. Executive Overview

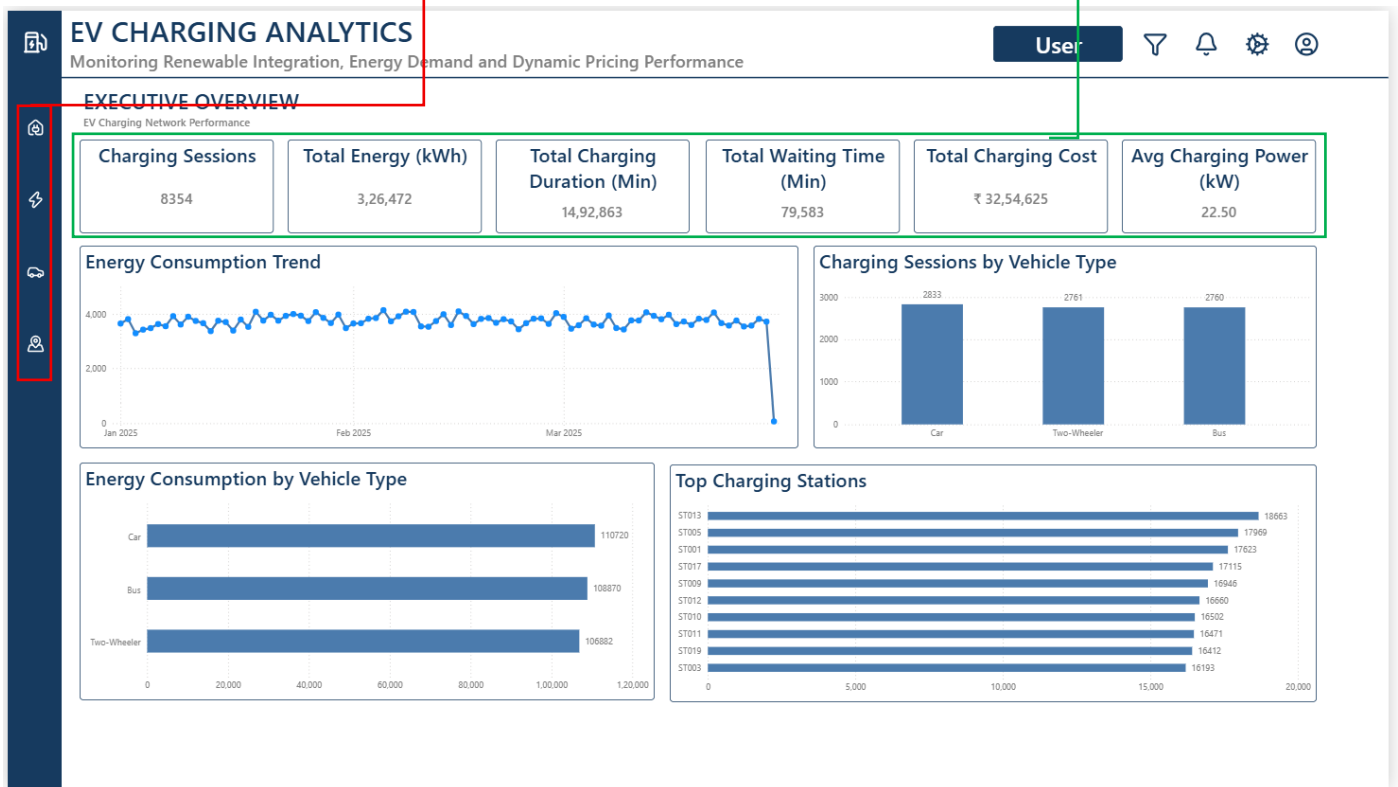
The **Executive Overview** is the main landing view.

It provides a high-level summary of the EV charging network.

The page contains six KPI cards and several analytical charts.

Main Sections

Six KPI Cards



### 4.1 KPI Cards

The six KPI cards are:

#### Charging Sessions

Shows the total number of charging sessions in the currently selected filter context.

#### Total Energy (kWh)

Shows the total energy consumed during charging sessions.

#### Total Charging Duration (Min)

Shows the combined charging duration across the selected data.

#### Total Waiting Time (Min)

Shows the total waiting time recorded across charging sessions.

#### Total Charging Cost

Shows the total charging cost.

#### Avg Charging Power (kW)

Shows the average charging power.

These KPIs automatically respond to the report filters.

## 5. Executive Overview Charts

### 5.1 Energy Consumption Trend

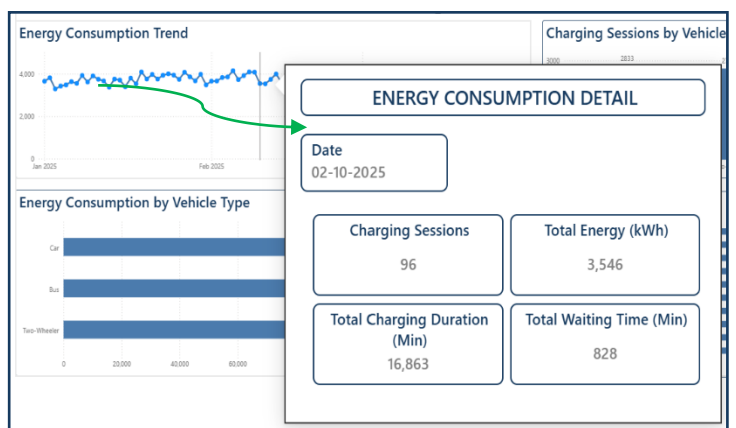
This line chart shows energy consumption over time.

#### How to use it

1. Move the mouse over a point on the line.
2. A tooltip appears.
3. The tooltip provides additional information for the selected date/context.

The chart can be used to identify:

- High-energy periods
- Low-energy periods
- Changes in charging demand
- Overall energy consumption patterns.



## 5.2 Charging Sessions by Vehicle Type

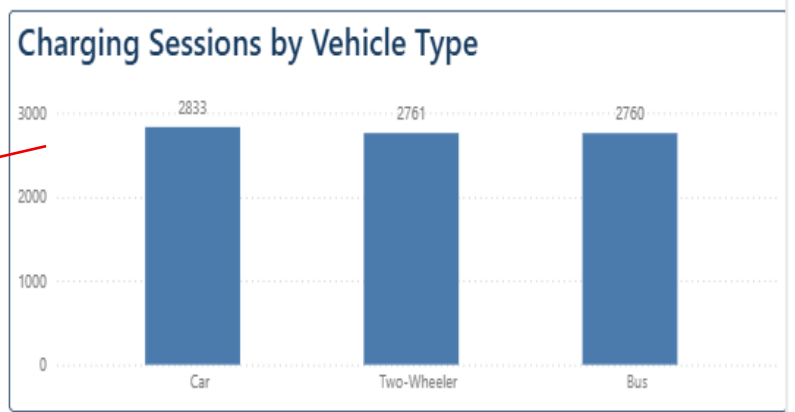
This chart compares charging sessions between vehicle categories.

The available vehicle categories include:

- Car
- Two-Wheeler
- Bus

Use this chart to understand which vehicle types contribute most to charging activity.

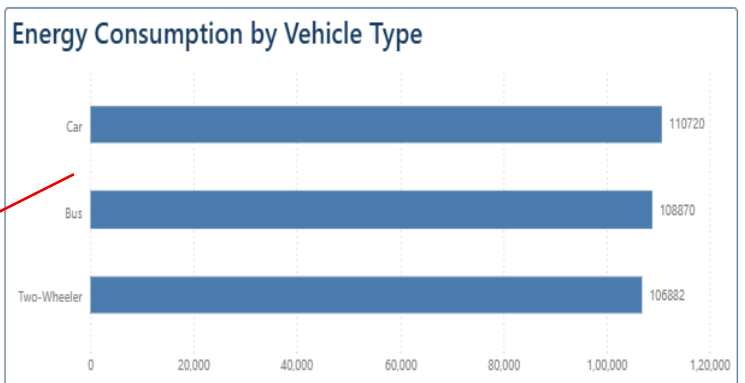
- Car — 2,833 sessions
- Two-Wheeler — 2,761 sessions
- Bus — 2,760 sessions



## 5.3 Energy Consumption by Vehicle Type

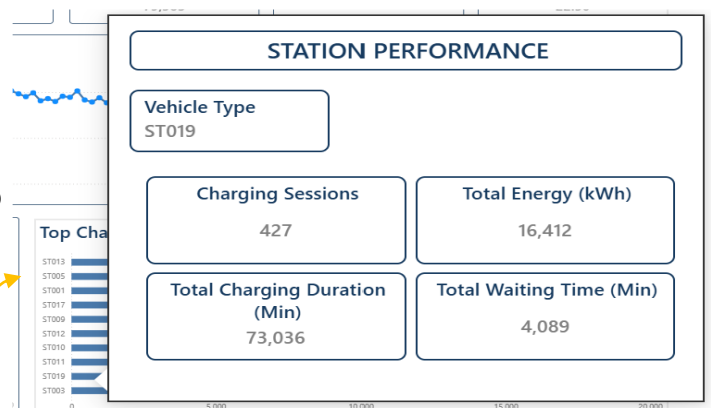
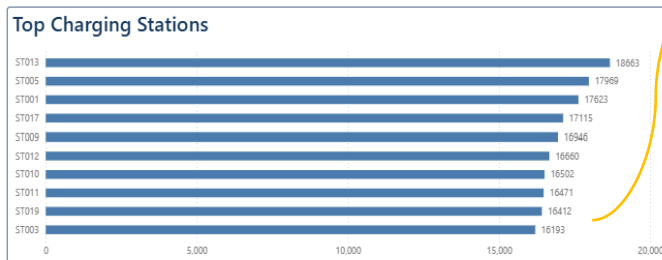
1. This chart compares total energy consumption across vehicle types.
2. It helps identify which vehicle category consumes the greatest amount of charging energy.

- Car — 110,720 kWh
- Two-Wheeler — 106,882 kWh
- Bus — 108,870 kWh



## 5.4 Top Charging Stations

1. This chart ranks charging stations according to their charging activity/energy performance.
2. Users can identify high-performing or high-demand stations.
3. This visual also supports the **Station Details tooltip** and can be used as a starting point for station-level drill-through analysis.



## 6. Using the Filter Panel

The dashboard contains a dedicated filter panel.

Click the **Filter/Funnel icon** in the top-right area.

The filter panel contains:

### Date

Allows the user to select a date range.

The current dataset covers:

**January 1, 2025 – March 29, 2025**

### Vehicle

Allows filtering by vehicle type.



Examples:

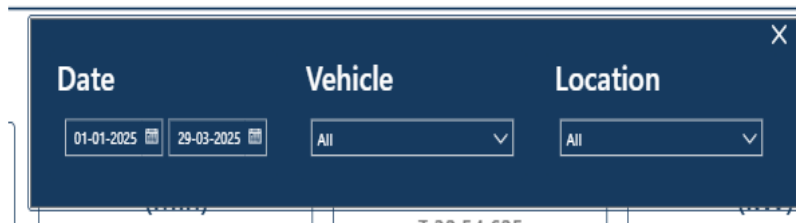
- Car
- Two-Wheeler
- Bus

### Location

Allows filtering by location type.

Examples include:

- Urban
- Highway



## 7. Applying Filters

To apply a filter:

1. Click the **Filter icon**.
2. Select the required Date range.
3. Select a Vehicle type if required.
4. Select a Location type if required.
5. Close the filter panel.

The selected filters affect the analytical visuals and KPI values throughout the report.

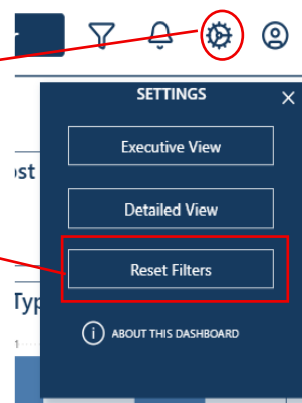
For example:

**Vehicle = Car**

will update the dashboard to show information associated with the selected vehicle type.

## 8. Resetting Filters

1. If filters have been applied and you want to return to the default dashboard state:
  - Open **Settings**.
  - Select **Reset Filters**.
2. Alternatively, use the Reset Filters option available inside the filter/settings workflow.
3. The dashboard returns to its default filter context.

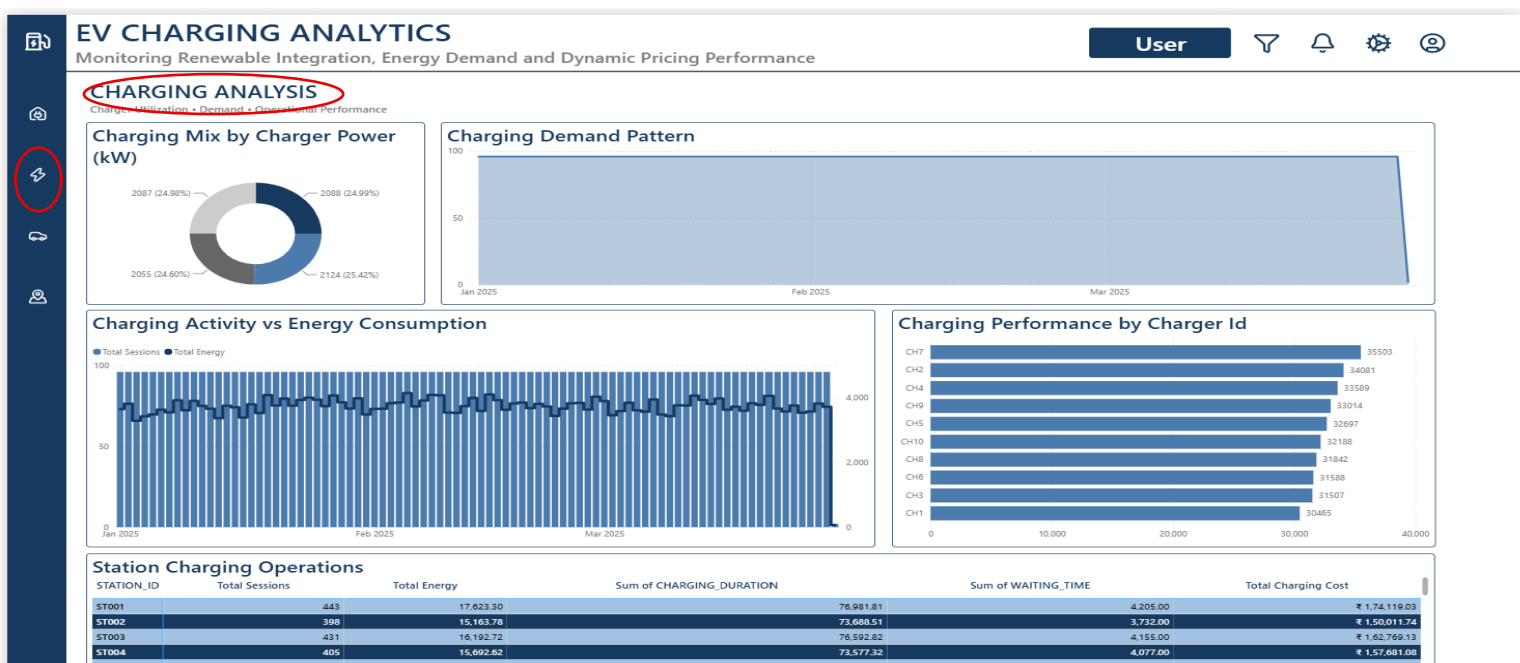


## 9. Charging Analysis

Click the **Charging/Lightning icon** on the left navigation bar.

This opens the **Charging Analysis** view.

This section focuses on charger utilization, charging demand, and operational performance.



- Total charging sessions
- Total energy consumption

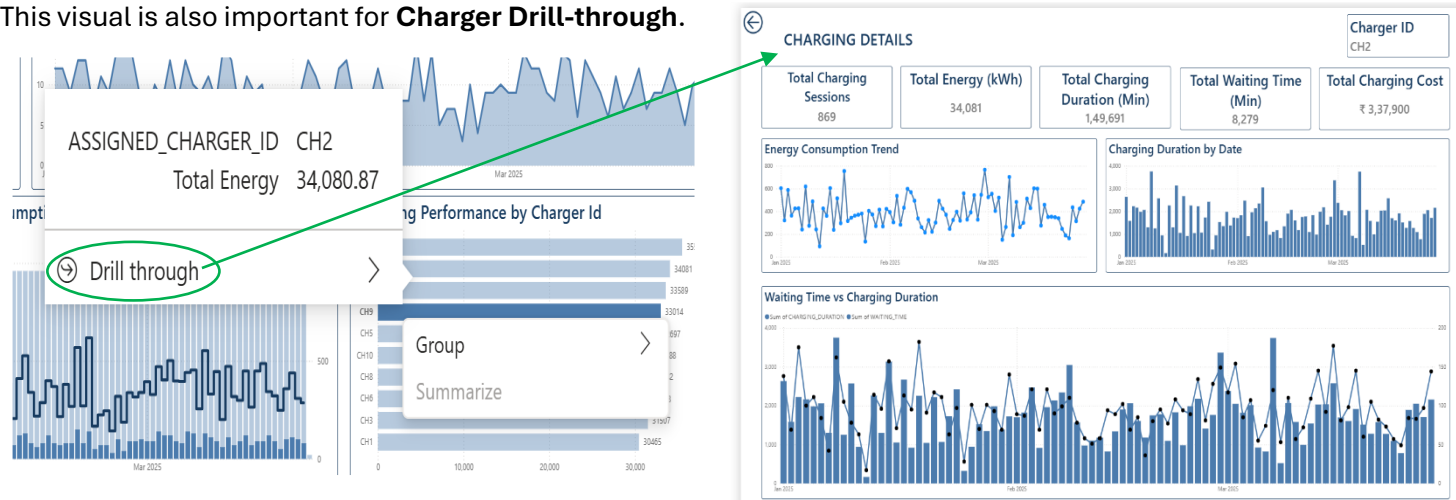
It allows users to compare charging activity against the amount of energy consumed.

#### 9.4 Charging Performance by Charger ID

This chart ranks chargers based on total energy performance.

The charger IDs can be used to identify high-performing chargers.

This visual is also important for **Charger Drill-through**.



#### 9.5 Station Charging Operations

The table provides station-level operational information.

The table contains information such as:

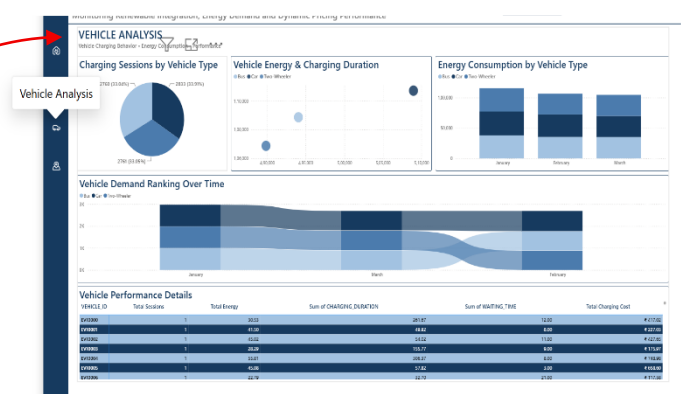
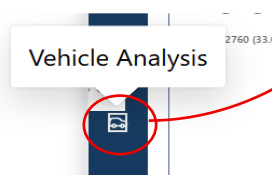
- Station ID
- Total Sessions
- Total Energy
- Charging Duration
- Waiting Time
- Total Charging Cost

| Station Charging Operations |                |              |                          |                     |                     |
|-----------------------------|----------------|--------------|--------------------------|---------------------|---------------------|
| STATION_ID                  | Total Sessions | Total Energy | Sum of CHARGING_DURATION | Sum of WAITING_TIME | Total Charging Cost |
| ST001                       | 443            | 17,623.30    | 76,981.81                | 4,205.00            | ₹ 1,74,119.03       |
| ST002                       | 398            | 15,163.78    | 73,688.51                | 3,732.00            | ₹ 1,50,011.74       |
| ST003                       | 431            | 16,192.72    | 76,592.82                | 4,155.00            | ₹ 1,62,769.13       |
| ST004                       | 405            | 15,692.62    | 73,577.32                | 4,077.00            | ₹ 1,57,681.08       |

Users can scroll through the table to inspect individual stations.

#### 10. Vehicle Analysis

1. Click the **Vehicle icon** on the left navigation bar.
2. This opens the **Vehicle Analysis** section.
3. This page focuses on vehicle charging behavior and energy consumption.

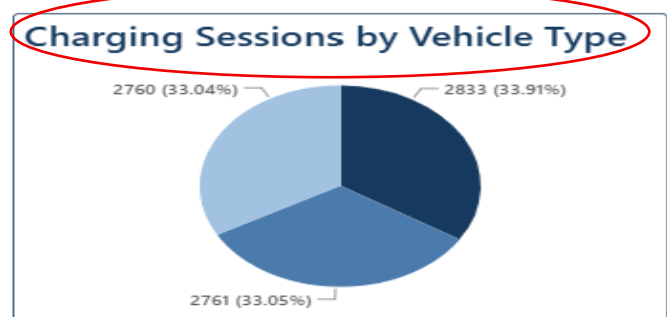


#### 10.1 Charging Sessions by Vehicle Type

The chart displays the distribution of charging sessions across:

- Car - 33.91%
- Two-Wheeler – 33.04%
- Bus – 33.05%

It provides a quick comparison of vehicle demand.



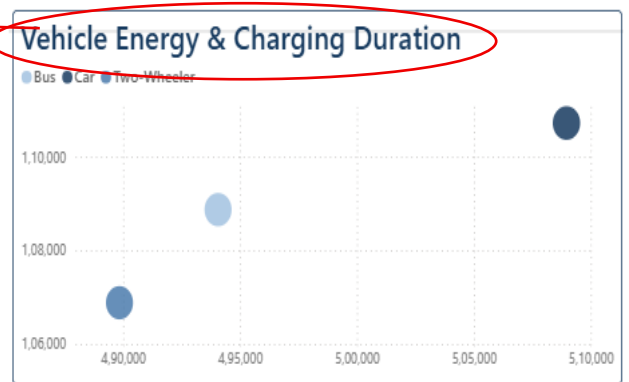
## 10.2 Vehicle Energy & Charging Duration

This scatter chart compares:

- Charging duration
- Energy consumption
- Vehicle type
- Charging session volume

The size of the data point represents charging-session activity.

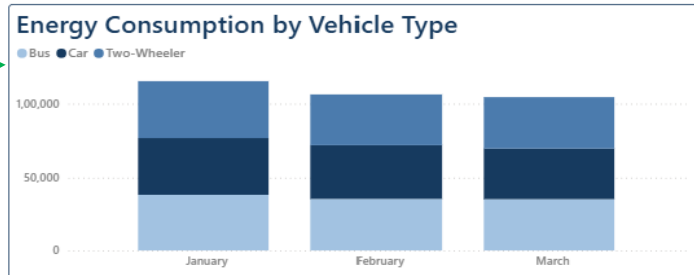
Use this visual to identify relationships between charging duration and energy consumption.



## 10.3 Energy Consumption by Vehicle Type

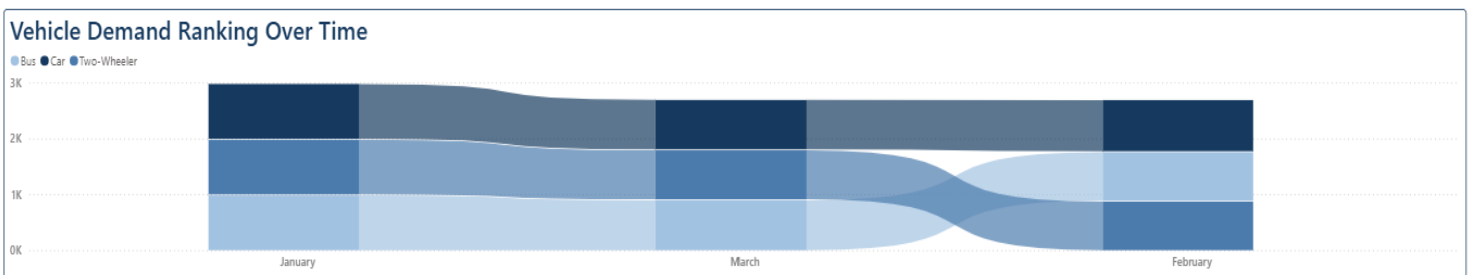
1. This chart shows energy consumption by vehicle type across different months.
2. It can be used to compare monthly energy consumption between:

- Car
- Bus
- Two-Wheeler



## 10.4 Vehicle Demand Ranking Over Time

1. This visual shows how vehicle-type demand changes over time.
2. It helps users understand changes in vehicle demand between different months.



## 10.5 Vehicle Performance Details

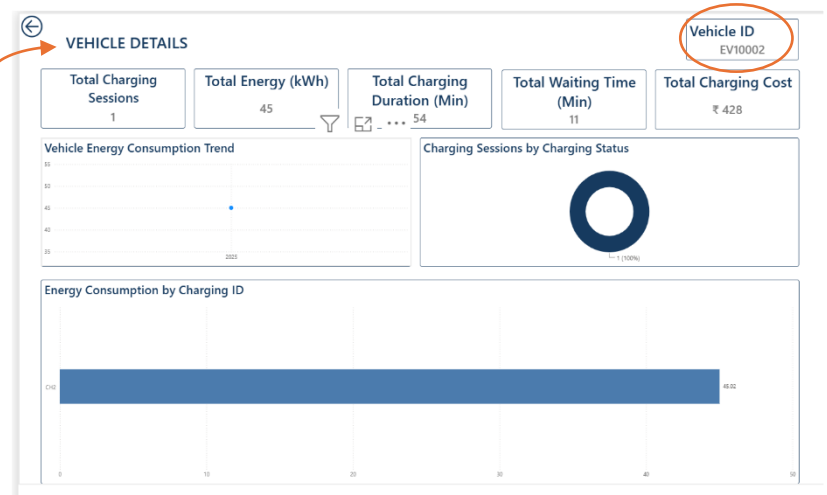
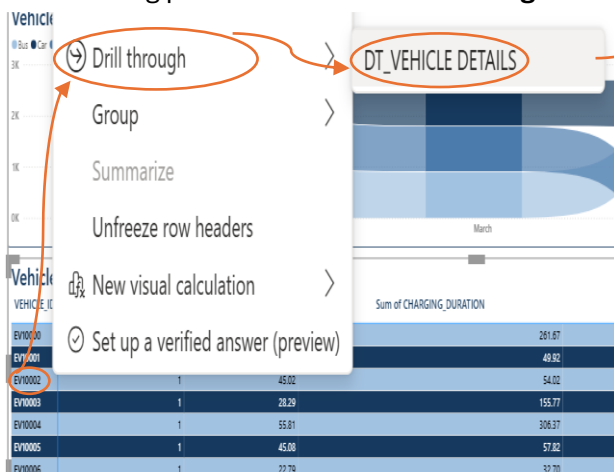
The table provides vehicle-level information.

The table includes:

- Vehicle ID
- Total Sessions
- Total Energy
- Charging Duration
- Waiting Time
- Total Charging Cost

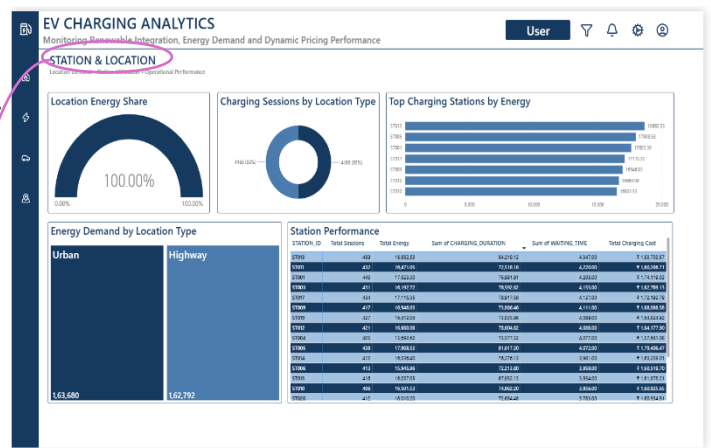
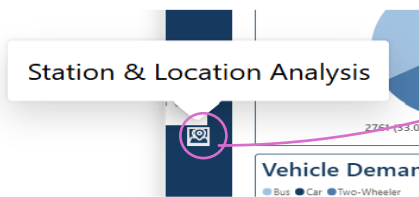
| Vehicle Performance Details |                |              |                          |                     |                     |
|-----------------------------|----------------|--------------|--------------------------|---------------------|---------------------|
| VEHICLE_ID                  | Total Sessions | Total Energy | Sum of CHARGING_DURATION | Sum of WAITING_TIME | Total Charging Cost |
| EV10000                     | 1              | 30.53        | 261.67                   | 12.00               | ₹ 417.02            |
| EV10001                     | 1              | 41.50        | 49.92                    | 8.00                | ₹ 227.03            |
| EV10002                     | 1              | 45.02        | 54.02                    | 11.00               | ₹ 427.65            |
| EV10003                     | 1              | 28.29        | 155.77                   | 9.00                | ₹ 175.97            |
| EV10004                     | 1              | 55.81        | 306.37                   | 8.00                | ₹ 748.98            |
| EV10005                     | 1              | 45.08        | 57.82                    | 3.00                | ₹ 658.60            |
| EV10006                     | 1              | 22.79        | 32.70                    | 21.00               | ₹ 117.58            |

This table is the primary starting point for **Vehicle Drill-through**.



## 11. Station & Location Analysis

1. Click the **Location/Map** icon on the left navigation bar.
2. This opens the **Station & Location Analysis** section.
3. This page focuses on station utilization and geographical/location-level performance.



### 11.1 Location Energy Share

1. This gauge shows the energy contribution associated with the selected location context.
2. It provides an overview of location-level energy distribution.



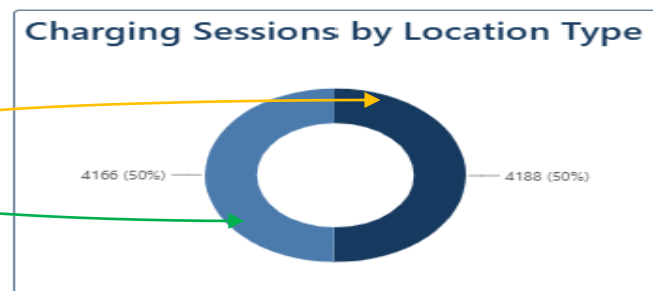
### 11.2 Charging Sessions by Location Type

1. This donut chart compares charging sessions by location type.

For example:

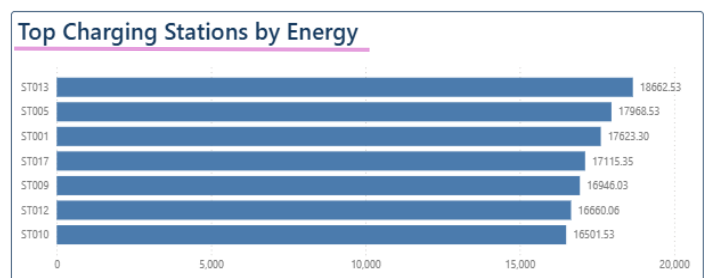
- Urban
- Highway

2. It helps identify where charging activity is concentrated.



### 11.3 Top Charging Stations by Energy

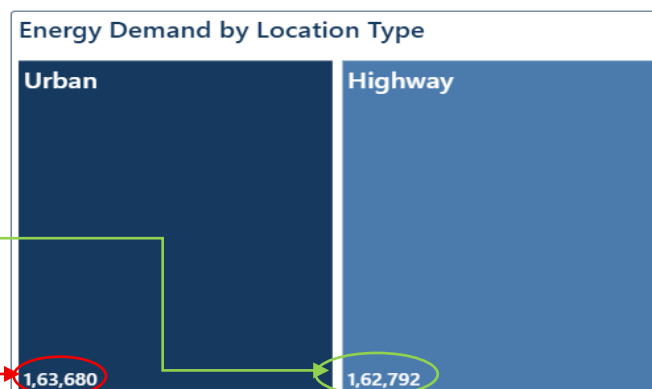
1. This horizontal bar chart ranks charging stations according to total energy consumption.
2. Use it to identify stations with the highest energy demand.



### 11.4 Energy Demand by Location Type

1. The treemap compares energy demand between location types.
2. Larger areas represent greater energy demand.

- Urban – 1,63,680
- Highway – 1,62,792





## 11.5 Station Performance

1. The station performance table provides detailed station-level information.
2. It contains:
  - Station ID
  - Total Sessions
  - Total Energy
  - Charging Duration
  - Waiting Time
  - Total Charging Cost

| Station Performance |                |              |                          |                     |                     |  |
|---------------------|----------------|--------------|--------------------------|---------------------|---------------------|--|
| STATION_ID          | Total Sessions | Total Energy | Sum of CHARGING_DURATION | Sum of WAITING_TIME | Total Charging Cost |  |
| ST013               | 459            | 18,662.53    | 84,219.12                | 4,347.00            | ₹ 1,83,703.57       |  |
| ST011               | 432            | 16,471.05    | 72,518.18                | 4,220.00            | ₹ 1,60,286.71       |  |
| ST001               | 443            | 17,623.30    | 76,981.81                | 4,205.00            | ₹ 1,74,119.03       |  |
| ST003               | 431            | 16,192.72    | 76,592.82                | 4,155.00            | ₹ 1,62,769.13       |  |
| ST017               | 434            | 17,115.35    | 78,917.58                | 4,127.00            | ₹ 1,72,192.79       |  |
| ST009               | 417            | 16,946.03    | 73,886.46                | 4,111.00            | ₹ 1,68,086.58       |  |
| ST019               | 427            | 16,412.04    | 73,035.86                | 4,089.00            | ₹ 1,64,824.92       |  |
| ST012               | 421            | 16,660.06    | 78,604.82                | 4,086.00            | ₹ 1,64,177.90       |  |
| ST004               | 405            | 15,692.62    | 73,577.32                | 4,077.00            | ₹ 1,57,681.08       |  |
| ST005               | 438            | 17,968.53    | 81,617.30                | 4,072.00            | ₹ 1,79,406.47       |  |
| ST014               | 412            | 16,036.40    | 76,276.13                | 3,981.00            | ₹ 1,63,259.01       |  |
| ST006               | 413            | 15,945.96    | 72,213.80                | 3,959.00            | ₹ 1,60,519.70       |  |
| ST015               | 416            | 16,057.05    | 67,852.13                | 3,954.00            | ₹ 1,61,878.21       |  |
| ST010               | 406            | 16,501.53    | 74,862.20                | 3,856.00            | ₹ 1,60,825.65       |  |
| ST020               | 412            | 16,010.20    | 72,694.48                | 3,783.00            | ₹ 1,60,934.51       |  |

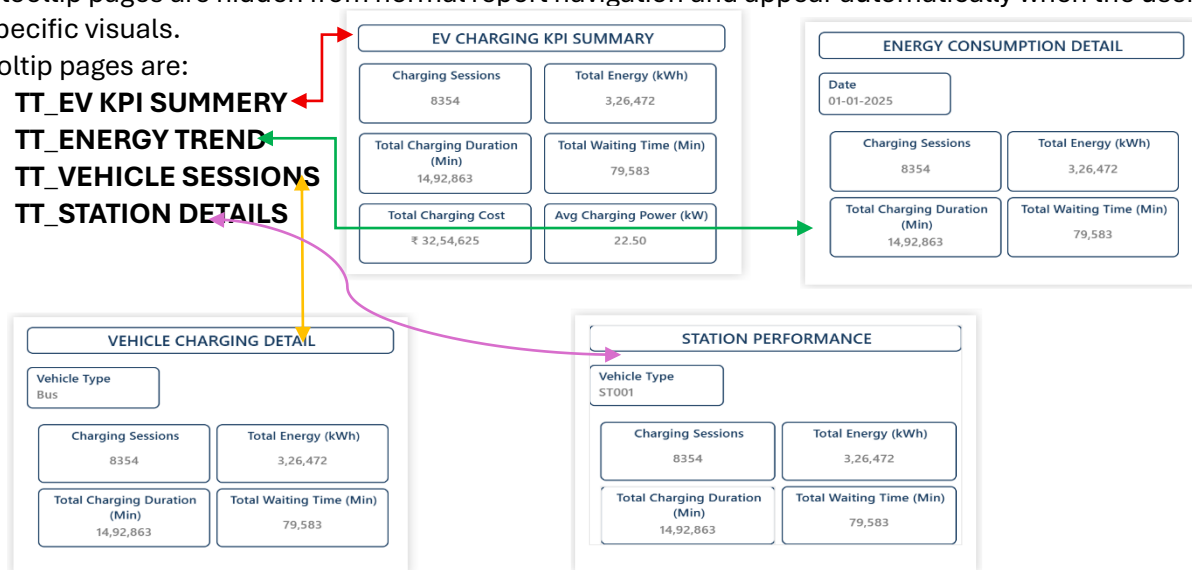
## 12. Interactive Tooltips

The dashboard contains **four dedicated tooltip pages**.

These tooltip pages are hidden from normal report navigation and appear automatically when the user hovers over specific visuals.

The tooltip pages are:

1. **TT\_EV KPI SUMMERY**
2. **TT\_ENERGY TREND**
3. **TT\_VEHICLE SESSIONS**
4. **TT\_STATION DETAILS**



## 13. KPI Tooltip

The KPI tooltip is available when hovering over the six Executive Overview KPI cards.

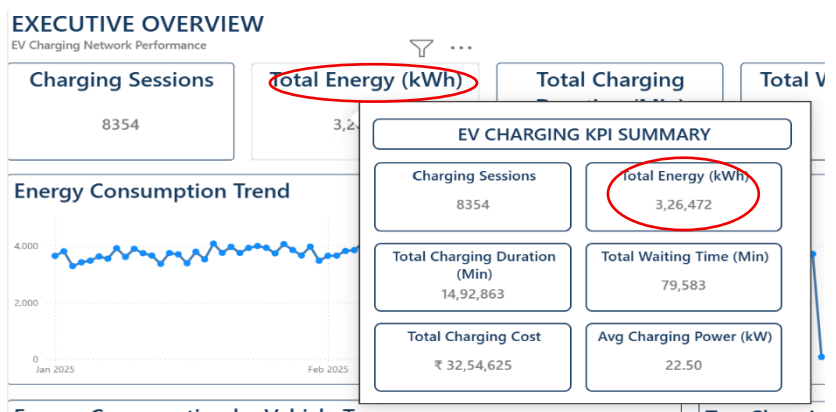
The tooltip provides contextual KPI information including:

- Charging Sessions
- Total Energy (kWh)
- Total Charging Duration (Min)
- Total Waiting Time (Min)
- Total Charging Cost
- Avg Charging Power (kW)

### How to use

1. Go to Executive Overview.
2. Move the mouse over any KPI card.
3. Wait for the tooltip to appear.
4. Review the additional KPI information.
5. Move the mouse away to close it.

The tooltip reflects the current filter context.



## 14. Drill-through Analysis

The report contains three dedicated drill-through pages.

### Drill-through pages

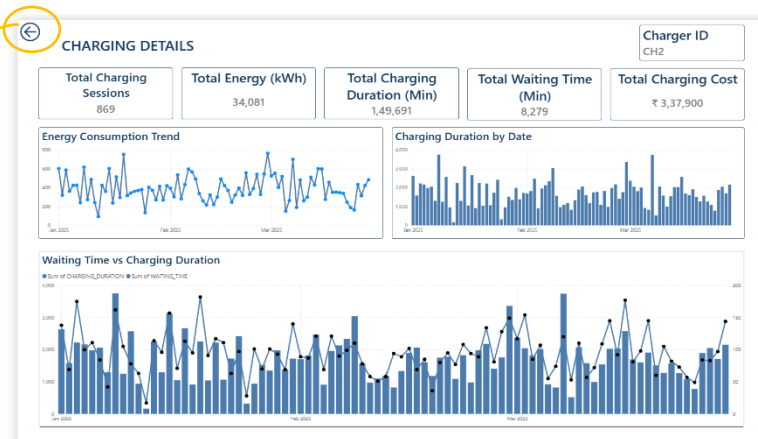
1. **DT\_CHARGING ANALYSIS**
2. **DT\_VEHICLE DETAILS**
3. **DT\_STATION DETAILS**

These pages are hidden from the normal navigation and are accessed through drill-through actions.



## 15. Returning from Drill-through

1. Every drill-through page contains a **Back** button.
2. After completing the detailed analysis:
  1. Click the **Back** button.
  2. Power BI returns to the previous report view.



## 16. Notifications

The dashboard contains a notification panel accessed using the **Bell icon**.

The current notification panel is **static**.

It contains:

### HIGH CHARGING

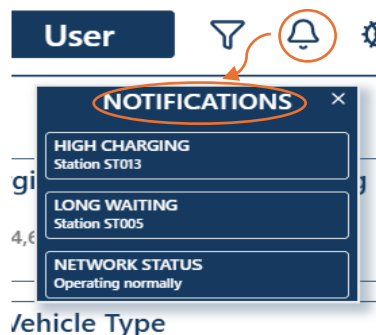
Station ST013

### LONG WAITING

Station ST005

### NETWORK STATUS

Operating normally



## 18. Settings Panel

Click the **Settings/Gear icon** to open the Settings panel.

The panel contains:

### Executive View

Returns the dashboard to the Executive Overview.

### Detailed View

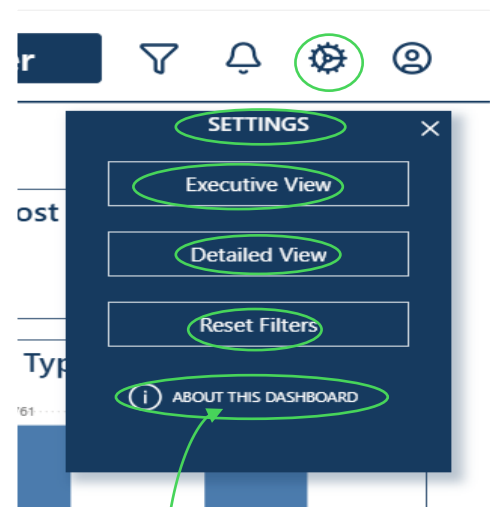
Opens the detailed Charging Analysis view.

### Reset Filters

Returns the dashboard to the default filter state.

### About This Dashboard

Opens information about the purpose and coverage of the dashboard.



## 19. About This Dashboard

The About panel explains the dashboard.

### Purpose

Analyze:

- Charging demand
- Energy consumption
- Operational efficiency
- Network performance

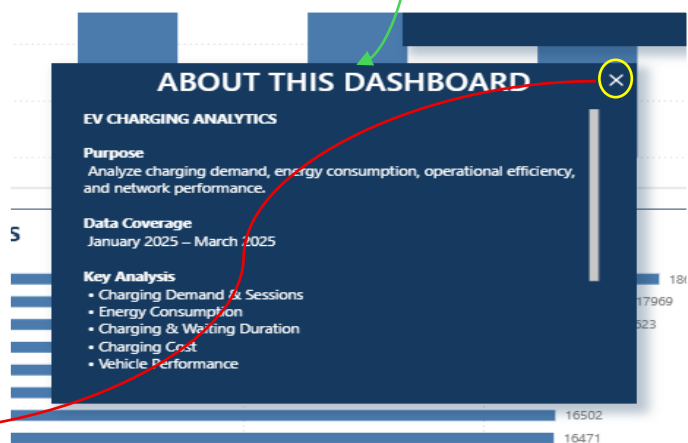
### Data Coverage

January 2025 – March 2025

### Key Analysis Areas

- Charging Demand & Sessions
- Energy Consumption
- Charging & Waiting Duration
- Charging Cost
- Vehicle Performance

Users can close the panel using the **X/Close icon**.



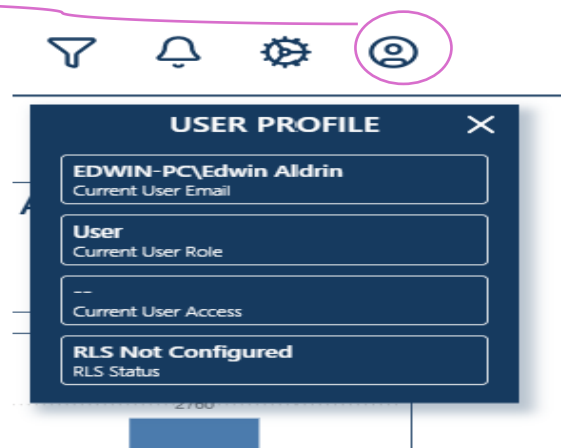
## 20. User Profile

Click the **User/Profile icon** in the top-right corner.

The User Profile panel displays information such as:

- Current User Email
- Current User Role
- Current User Access
- RLS Status

The panel confirms whether **Row-Level Security (RLS)**, is enabled for the current user.



## 21. Row-Level Security (RLS)

The dashboard uses RLS to control what data a user can access.

The User Profile panel provides an indication of the current user's security context.

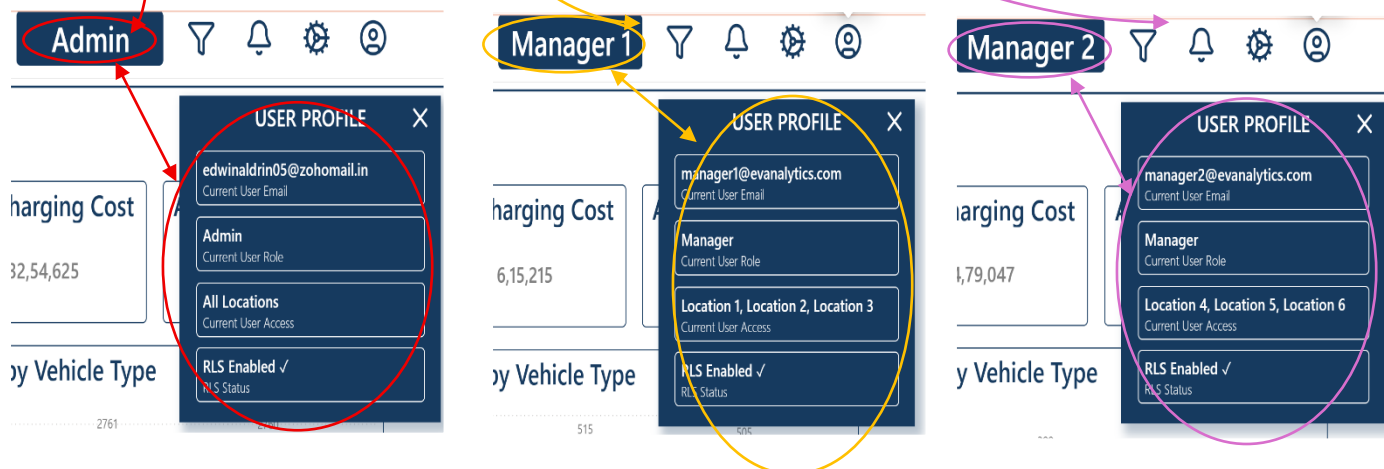
For example, an administrator may see:

**Role:** Admin, Manager 1, Manager 2

**Access:** All Locations, (1,2,3), (4,5,6)

**RLS:** Enabled

A normal user may see a different role and a restricted location/data scope depending on the security configuration.



\*\*\* continue to next page for Complete Dashboard Usage Flow \*\*\*



## 21. COMPLETE DASHBOARD USAGE FLOW



**Note:** The notification panel (Bell icon) shows static operational information and does not change based on filters or current data.